

What was tried, what failed, what changed.

Four pivots from default assumption to measured operator. Each pivot replaced an unverified habit with a measurement.

Exp A · Ch. 6

Autonomous spectral radius → driven Lyapunov exponent

Tried

Cite $\rho(W) < 1$ (the field's default for ESN stability) as evidence the reservoir is in the usable regime.

Failed

$\rho(W)$ is an autonomous, undriven property of W . It says nothing about how the reservoir behaves under real ERP input. $\rho(W) < 1$ does NOT imply the reservoir contracts along driven trajectories.

Changed

Compute the driven Lyapunov exponent λ_1 via the Benettin algorithm along 3,165 real ERP trajectories. Stability is now a MEASURED property of the system in operation, not a cited property of W in isolation.

$\rho(W) = 0.2647549015$ (autonomous); driven λ_1 measured on 3,165 trials.

Exp C · Ch. 6

Hyperparameter chosen a priori → operating-regime measurement

Tried

Set leak rate $\beta = 0.05$ from a biological time-constant argument and use it without measuring the reservoir's memory-capacity response across β .

Failed

Measured MC peak is at $\beta^* \approx 0.012$ ($MC \approx 0.835$), NOT at the dissertation's chosen $\beta = 0.05$ ($MC \approx 0.763 \approx 91\%$ of peak). The 'a priori' choice does not maximize measured capacity.

Changed

Build the Propagation Operating Characteristic — a measured MC curve over β . Re-anchor $\beta = 0.05$ inside the measured plateau ($0.010 \leq \beta \leq 0.118$ all give $MC \geq 0.75$) under a JOINT constraint: $MC \geq 90\%$ of peak AND $1/\beta$ matches the α -band period.

~~β is not chosen by validation accuracy.~~
 $\beta = 0.05 \rightarrow MC \approx 0.763$ (91% of measured peak). Plateau: $\beta \in [0.010, 0.118]$.

Exp D · Ch. 5

Single global channel permutation → per-trial channel permutation

Tried

Test the spatial-feature claim with a single global channel permutation π applied to every trial (or with parametric p-values that assume EEG-noise structure). [author confirm]

Failed

A flat linear classifier compensates for a single π by learning new weights — cross-validated AUC stays unchanged. The 'wrong null' does not actually test what the dissertation claims about spatial structure.

Changed

Generate an independent π_i PER TRIAL. The classifier now sees a (channel_pos, feature_idx) cell that contains data from a DIFFERENT EEG channel on every trial. Per-channel signal is destroyed; only per-trial overall power survives. The STRICTEST possible spatial null.

500 perms × 5 CV folds. Observed AUC sits well outside the null distribution per class.

Cross-cutting · Ch. 4-5

Trial-level cross-validation → subject-level StratifiedGroupKFold

Tried

Score classifiers with trial-level k-fold cross-validation — the default in many clinical-EEG papers. [author confirm: was this an earlier audited choice in the dissertation, or always subject-level?]

Failed

Trial-level CV in clinical EEG inflates accuracy by 10–15 percentage points by leaking same-subject correlation across train and test folds. The inflated number does not generalize to unseen subjects.

Changed

Subject-level StratifiedGroupKFold across every classification claim in the dissertation. Each subject's trials live entirely in train OR test, never both. Trial-level numbers were never the headline.

Documented in Exp D companion notes §3.3 as the place to disclose the audit.

Each pivot replaced an unverified assumption with a measurement. The deck's defensibility lives here.